

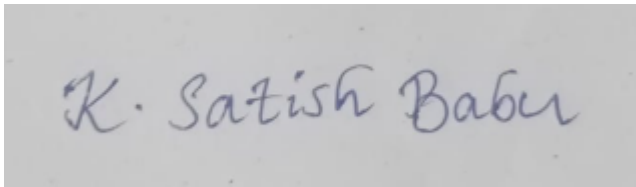
Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

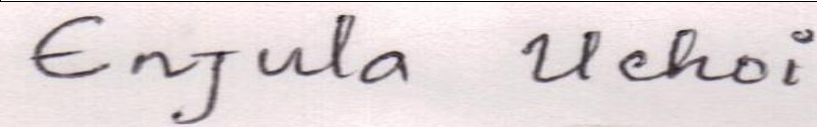
Details of Invention for better understanding:

1. TITLE: Smart Sight Helmet

2. INTERNAL INVENTOR(S)/ STUDENT(S): All fields in this column are mandatory to be filled

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	Signature (Mandatory)	

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	Signature (Mandatory)	

For External Inventors, NOC (No Objection Certificate) from the affiliated institute/university/Industry/lab etc. is mandatory for each individual inventor and their respective topic. For NOC, format is attached below.

(FOR ADDITIONAL INVENTORS, PLEASE ADD ROWS)

2. **ABSTRACT:**

This is where our Smart AI Helmet comes in. It is an intelligent helmet designed to improve road safety. It is compact and light in weight and is specifically designed for bike riders.

The Smart AI Helmet is integrated with sensors and artificial intelligence. It uses these sensors to monitor the speed of your bike. It also detects dangerous activities and immediately alerts you. This ensures that you stay on track and avoid accidents.

The sensors of our Smart AI Helmet are connected to a powerful processing unit. This unit uses artificial intelligence to analyze the data collected by the sensors. It is also connected to a mobile app. This app works as a safety assistant for bike riders. It offers detailed insights into your activities and alerts you if any dangerous activity is detected.

The most important feature of our Smart AI Helmet is its accident detection system. This system sends alerts to nearby hospitals and police stations in case of an accident. This is done with the help of GPS location tracking.

The Smart AI Helmet offers various advantages to bike riders. It ensures safer rides and reduces the risk of accidents. It also offers quick assistance during emergency situations.

3. DESCRIPTION OF THE INVENTION:

Introducing the Smart AI Helmet, a smart safety device developed specifically for bike riders. The helmet functions as a personal riding assistant, helping to avoid accidents and ensure prompt response during emergency situations.

The helmet includes sensors to monitor the bike's speed and detect sudden movements or impacts. These sensors are connected to a microcontroller such as ESP32. The microcontroller uses AI to analyze the information and detect risky riding behaviors such as oversteering. The helmet immediately alerts the rider when such risky riding behaviors are identified.

The device is wirelessly connected to a mobile application. The mobile application displays the riding information to the user in a simple format. The mobile application sends notifications to the rider when unsafe riding behaviors are identified.

In the event of an accident, the helmet detects the accident and sends alerts along with the GPS location of the rider to hospitals and police stations. The helmet can also send alerts to emergency contacts.

The Smart AI Helmet ensures road safety by reducing accident risks and providing prompt response during emergency situations.

A. PROBLEM ADDRESSED BY THE INVENTION:

The Smart AI Helmet aims to solve the problem of road accidents and emergency response being delayed for bike riders. Current helmets protect the rider physically but do nothing to help prevent accidents or respond to emergencies effectively. Many accidents happen due to over-speeding, lack of awareness, or unexpected changes in road conditions. Moreover, the victims are unable to receive timely medical aid because no one is notified about the emergency.

The invention aims to solve these problems with the help of advanced sensors and artificial intelligence incorporated into a compact helmet that a bike rider can wear. It monitors the speed of the bike continuously and also detects unsafe bike riding patterns such as over-speeding or accidents. The data is processed using a microcontroller that sends instant alerts to warn the rider before any accidents happen.

The Smart AI Helmet also has a facility for detecting accidents automatically. When a bike crashes, it sends alerts with the rider's GPS location to nearby hospitals and police stations for assistance.

The helmet also sends alerts to family members for emergency response. Moreover, the helmet is connected to a mobile app that monitors bike riding patterns and sends useful safety information for effective response.

B. OBJECTIVE OF THE INVENTION:

1. The main aim is to design a smart, wearable, and user-friendly helmet that improves the safety of the rider through continuous monitoring of riding conditions such as speed and sudden movements.
2. The invention also aims to improve the safety of the rider through the detection of dangerous riding conditions such as overspeeding, which will send alerts through artificial intelligence, thus enabling the rider to make immediate corrections and prevent accidents.
3. Another main aim of this invention is to design an integrated emergency alert system that will detect accidents and send immediate alerts along with the location via GPS, thus alerting nearby hospitals, police stations, and emergency contacts.
4. The invention also aims to design a mobile application that will help users track riding patterns, provide safety insights, and improve riding habits through immediate feedback and alerts.
5. Through the improvement of accident prevention and the provision of immediate emergency services, the invention aims to reduce accidents and improve safety and confidence for bike riders.

C. STATE OF THE ART/ RESEARCH GAP/NOVELTY: Describe your invention fulfil the research gap? Mimimum 5 papers

Sr. No.	Patent I'd	Abstract	Research Gap	Novelty
1.	IN202241072092	IoT-based helmet that uses sensors to detect accidents, provides navigation, and sends accident alerts to emergency services.	Focuses solely on accident detection and sending alerts after accidents	Integration of IoT with navigation and accident alerts.

2.	WO2023156789A1 (WIPO)	Intelligent helmet with a gyroscope + accelerometer for detecting falls and sending emergency notifications using GPS + GSM/4G networks.	Reactive design; doesn't provide safety measures for accidents that may happen due to high speed.	IoT for automatic emergency notification.

3.	IN202341041234 (India)	Smart helmet that uses ESP32, MPU6050 accelerometer, GPS, and GSM for detecting accidents and sending SMS for the same.	Limited to detecting accidents; doesn't monitor speed or send alerts for the same.	Low-cost IoT hardware for accident alerts.
4.	US20190234567A1 (USA)	Helmet with a communication system, GPS tracking, and accident detection for sending alerts to emergency services or family members.	Focuses solely on sending alerts; doesn't provide safety measures for accidents that may happen due to high speed.	Combination of safety measures with communication .

D. DETAILED DESCRIPTION:

The Smart AI Helmet can be defined as a sophisticated, lightweight headgear that mimics other conventional safety gear while incorporating advanced technologies for improving rider safety in a smarter way. The helmet is intended for comfortable wear on the rider's head, monitoring the conditions of riding without causing any hindrance in practice.

Inside the helmet, a set of sensors are embedded for monitoring vehicle speeds, abrupt movements, and impacts. These sensors are highly sensitive, allowing the detection of minute variations in movement or unusual riding patterns. The information collected from these sensors is processed by a powerful microcontroller, ESP32, which uses artificial intelligence for monitoring the behavior of the rider and recognizing possible dangers.

The helmet also maintains a wireless connection with a smartphone application, which acts as a safety tool for riders, providing information in a clear and concise manner regarding speeds, patterns, and safety concerns. Riders can also track their behavior and evaluate the safety of driving.

If the system detects overspeeding or other unsafe conditions, immediate alerts are sent through the helmet or the smartphone application, allowing the rider to take corrective actions immediately. In case of an accident, the helmet also detects the impact and sends out emergency alerts along with the location of the rider, as identified through the global positioning system, to nearby hospitals, law enforcement agencies, and other emergency contacts.

Therefore, the Smart AI Helmet can be defined as a safety gear for riders, which goes beyond conventional safety gear by incorporating advanced technologies for improving safety in a smarter way, thereby preventing accidents, responding to emergencies, and ensuring safe riding and security for riders.

E. RESULTS AND ADVANTAGES:

The ****Smart AI Helmet**** is an innovative invention that offers a wide array of benefits for the improvement of road safety and emergency response for bike riders. This helmet allows for the advanced monitoring of the riding conditions and the early detection of unsafe riding practices. This enables the riders

to take necessary precautions to prevent accidents.

One of the primary advantages of the **Smart AI Helmet** is the provision of real-time notifications for unsafe practices such as overspeeding. This enables the riders to take necessary precautions to prevent accidents. The accident detection system of the helmet enables the emergency response system to immediately notify nearby hospitals and police stations in the case of an accident. This saves a lot of time and enables the provision of medical assistance.

Moreover, the system can be connected to a mobile application for the improvement of the user interface. This application enables the user to understand their riding patterns and improve their practices over time. Moreover, the system can provide instant notifications and updates to the user.

Furthermore, the system can improve the communication response for the riders. This can be done by notifying the family members in the case of an accident. In summary, the **Smart AI Helmet** can improve the safety of the riders and prevent accidents. Moreover, the system can enable the provision of quick response for emergency situations and save lives.

F. EXPANSION:

The efficiency of the Smart AI Helmet can be further enhanced through a series of developments and improvements, thus making it a more comprehensive safety gear for riders:

1) Advanced Sensor Integration: The Smart AI Helmet can be further enhanced by the addition of other sensors such as environmental sensors (air quality and weather conditions), fatigue sensors, and obstacle sensors to make it a more comprehensive safety gear.

2) Artificial Intelligence Enhancement: The helmet can be further improved by making its artificial intelligence algorithms more capable of predicting possible accidents by analyzing riding patterns, traffic, and road conditions. The helmet can also be improved to provide riders with personal safety recommendations to make informed decisions regarding road safety.

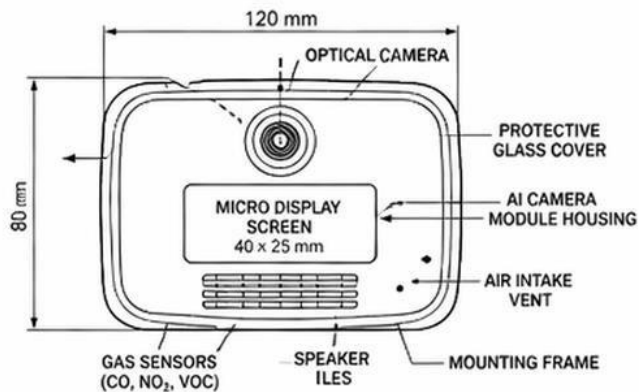
3) Integration with Smart Vehicles and Infrastructure: The helmet can be improved by making it capable of interacting with smart traffic systems, GPS navigation, and other vehicles to access information regarding traffic conditions and road safety.

4) Partnership with Emergency Services: The helmet can be improved by making it a partner with hospitals, police authorities, and other emergency services to make rescue operations more efficient and prompt.

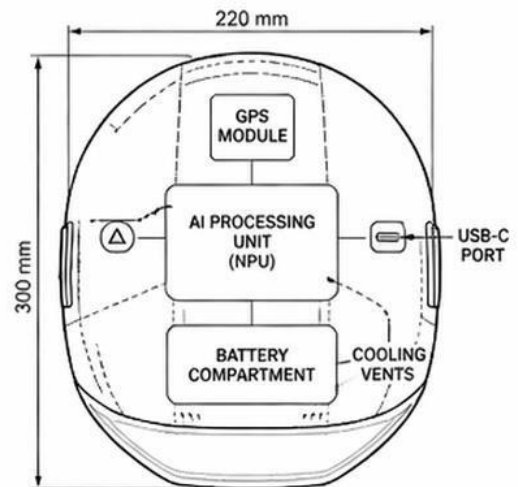
5) Community Safety Network: The helmet can be improved by making it a mobile platform through which riders can share information regarding road safety and accident hotspots, thus making it a part of a larger community network working towards road safety.

6) Continuous Improvement: The helmet can be further improved by making it a part of a series of developments and innovations, thus making it a more efficient and comprehensive safety gear. These developments and improvements have the potential to make a huge impact on road safety.

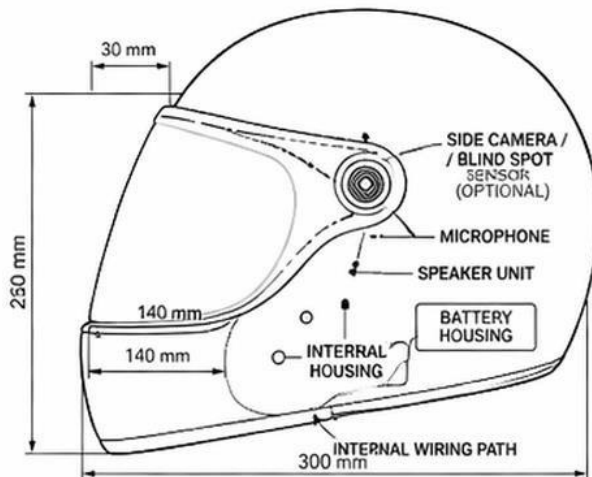
G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:



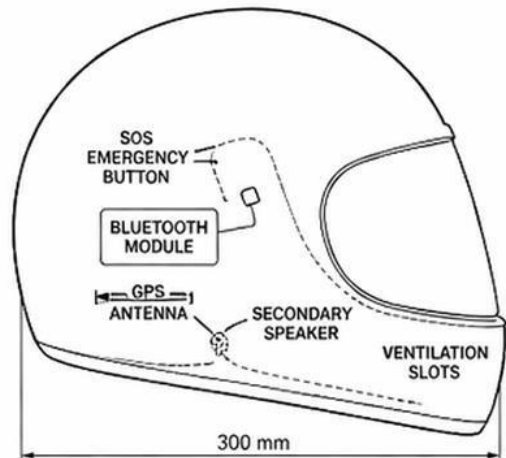
FRONT VIEW



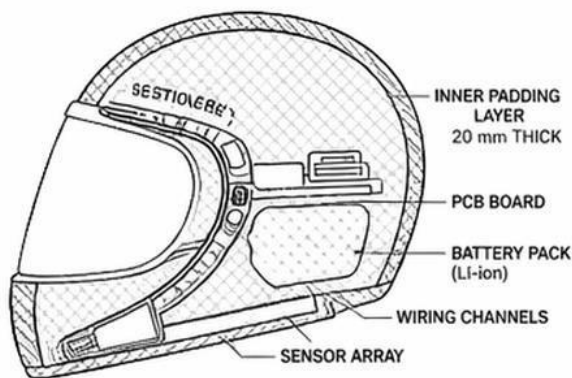
TOP VIEW



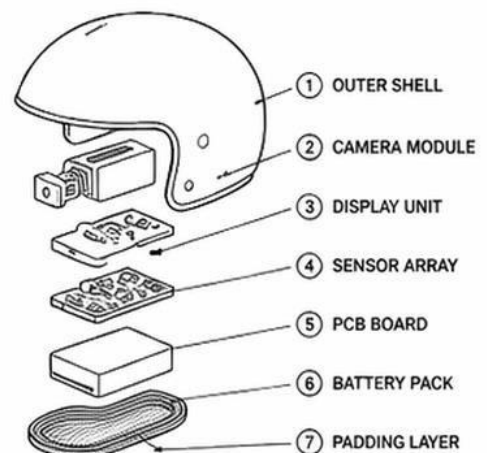
RIGHT SIDE VIEW



LEFT SIDE VIEW



SECTIONAL VIEW A-A



EXPLODED VIEW

AI-BASED SMART RISK DETECTION HELMET FOR TWO-WHEELER SAFETY

DRAWN BY: _____

SCALE: 1:2

UNITS: mm

SHEET NO: 1/5

G. EXISTING DATA: Any clinical or comparative data necessary enough to support your invention. (Comparative)

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

A. Have you described or shown your invention/ design to anyone or in any conference?	YES ()	NO (✓)
B. Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing your invention)?	YES ()	NO (✓)
C. Has your invention been described in any printed publication, or any other form of media, such as the Internet?	YES ()	NO (✓)
D. Do you have any collaboration with any other institute or organization on the same? Provide name and other details.	YES ()	NO (2713)
E. Name of Regulatory body or any other approvals if required.	YES (✓)	NO ()

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us.

Not available

6. Provide the terms and conditions of the MOU also if the work is done in collaboration within or outside university (Any Industry, other Universities, or any other entity).

No

7. Potential Chances of Commercialization.

The Smart AI Helmet is a product of innovative technological advancement and has a huge market potential for commercialization within the road safety and smart transport market. In addition, within the context of increasing road accident statistics globally, there is a growing need to develop road safety products beyond traditional protective gear. The product not only offers protection but also comes equipped with intelligent features such as accident prediction and automatic notification of medical centers and authorities. These features make the product highly attractive to consumers. The product is user-friendly and has a wide market appeal to various categories of consumers, including ordinary road users, courier services, and travelers. The product can send automatic notifications to medical centers and authorities of a road accident, thus greatly increasing its market potential. The product has huge market potential within the business and commercial sector, including logistics companies, ride-hailing services, and fleet management systems. The product is highly relevant to the business sector since road safety is a major concern. The government is actively promoting road safety and smart city development. Overall, the product has huge market potential and is highly attractive due to its features and the growing demand.

8. can be contacted for commercialization along with the website link.

1. **Bosch Mobility Solutions** – <https://www.bosch-mobility.com/Medtronic> -
2. **Hero MotoCorp** – Omron Healthcare - <https://www.omronhealthcare.com/>
3. **TVS Motor Company** – <https://www.tvsmotor.com/>
4. **Studds Accessories Ltd.** – <https://www.studds.com/>
5. **Steelbird Helmets** – <https://www.steelbirdhelmet.com/>
9. Any basic patent which has been used and we need to pay royalty to them.

No

10. **FILING OPTIONS:** Please indicate the level of your work which can be considered for provisional/ complete/ PCT filings (Mandatory to mention)

Provisional Design

11. KEYWORDS: Smart Helmet, Accident Detection, AI-based Safety, Rider Safety, Speed Monitoring, Real-time Analysis, Crash Detection System, GPS Tracking, Emergency Alert System, Intelligent Transportation, Riding Behavior Analysis, IoT-based Helmet

(Letter Head of the external organization)

NO OBJECTION CERTIFICATE

This is to certify that I (Name of the external person) have no financial assistance in filing any patent form from Lovely Professional University.

We have no objection if Lovely Professional University files any patent with the name of our employee (name of the external person) as co-inventor with (Names of LPU faculty/Staff and all co-inventors) having title as per the patent idea request file (LPU Idea Request I'D....).

Further, our institution will not raise any objections later concerning the filing and commercialization of the said patent.

(Authorised Signatory)